

Streptococcal Toxic Shock Syndrome (STSS): A Comprehensive Disease Characteristics Report

Disease: Streptococcal Toxic Shock Syndrome **MONDO ID:** MONDO:0020544 **Category:** Infectious Disease **Report type:** Aggregated disease-level synthesis (literature-derived; not derived from individual EHR records)

Summary

Streptococcal Toxic Shock Syndrome (STSS) is an acute, toxin-mediated, fulminant complication of invasive Group A *Streptococcus* (GAS; *Streptococcus pyogenes*) infection — and, less commonly, of group C/G streptococci (*Streptococcus dysgalactiae* subsp. *equisimilis*). It is defined clinically by the triad of **high fever, hypotensive shock, and multi-organ dysfunction**, frequently accompanied by a "sunburn-like" erythematous rash with subsequent desquamation and, in a large fraction of cases, necrotizing soft-tissue infection (necrotizing fasciitis). The core mechanism is not genetic in the patient but microbial and immunological: secreted **superantigens** (streptococcal pyrogenic exotoxins SPE-A/SPE-C, SSA, SMEZ) cross-link MHC class II molecules on antigen-presenting cells directly to the V β region of the T-cell receptor, bypassing conventional antigen processing and activating up to ~20% of the entire T-cell repertoire. This triggers a massive polyclonal **cytokine storm** (TNF, IL-1, IL-2, IL-6, IFN- γ , IL-17), which — together with streptolysin O (SLO)-mediated endothelial cytolysis — produces capillary leak, distributive shock, disseminated intravascular coagulation (DIC), and cascading failure of kidneys, liver, lungs, heart, and the hematologic system.

Although STSS is not a heritable disorder, the **host HLA class II genotype is the principal determinant of disease severity**: the same GAS clone can cause severe or mild invasive disease depending on how the patient's class II haplotype presents the superantigens. Epidemiologically, STSS accounts for roughly 20–24% of invasive GAS (iGAS) infections and is consistently the strongest independent predictor of iGAS mortality. The emergence and clonal expansion of the hyper-toxigenic **emm1/M1UK (M1T1/ST28) lineage**, which over-produces

SPE-A, has driven post-COVID increases in invasive disease and STSS. Risk is age-stratified — varicella and pharyngotonsillitis (with NSAID use as a key gene–environment interaction) in children; malignancy, diabetes, IV drug use, chronic wounds, and age >65 in adults.

Management rests on a three-pillar strategy: **early aggressive surgical source control** (debridement of necrotizing infection, potentially amputation), **combination antibiotics** (a β -lactam for bactericidal killing PLUS a protein-synthesis inhibitor — clindamycin or linezolid — to suppress toxin production), and **adjunctive intravenous immunoglobulin (IVIG)** plus intensive organ support. Despite this, STSS carries the highest mortality of all invasive GAS syndromes (~20–45% in dedicated cohorts), and **no licensed GAS vaccine exists**. This report synthesizes 14 confirmed findings across all 15 knowledge-base sections, grounded in 66 reviewed papers.

Key Findings

F001 — STSS is a superantigen-driven fulminant manifestation of invasive GAS

STSS is defined clinically by the triad of high fever, hypotensive shock, and multi-organ dysfunction, caused by GAS toxins. Superantigens (SPE-A, *speC*, *speG*) cross-link MHC class II and TCR V β , driving polyclonal T-cell activation and a cytokine storm; streptolysin O (SLO)-mediated endothelial lysis contributes to capillary leak and shock. As stated in a 2025 case report, *"The STSS triad encompasses high fever, hypotensive shock, and a 'sunburn-like' rash with desquamation. STSS, like Toxic Shock Syndrome (TSS), is a rare complication of streptococcal infections caused by Group A"* [PMID: 39886786](#). A genomically characterized fatal case links the specific virulence factors mechanistically: *"The slo-mediated endothelial lysis and speC/G-induced cytokine storm contributed to capillary leakage and rapid clinical deterioration"* [PMID: 41714973](#).

Ontology anchors: MONDO:0020544 (STSS); NCBITaxon:1314 (*S. pyogenes*); protein superantigens SPE-A, SPE-C.

F002 — Host HLA class II genotype determines severity of invasive GAS/STSS

The same GAS clone can cause severe or non-severe invasive disease depending largely on the patient's HLA class II type. Certain haplotypes present streptococcal superantigens in a way that elicits potent inflammatory responses leading to organ failure and shock, while others confer relative protection. The landmark review states: *"Certain class II haplotypes present the streptococcal superantigens in a way that results in responses, whereas others present the same superantigens in a way that elicits very potent inflammatory responses that can lead to organ failure and shock"* [PMID: 14620152](#). This is the closest thing to a "genetic risk factor" for STSS — a host-side modifier of an infectious disease, not a Mendelian causal locus.

F003 — Clindamycin and IVIG are associated with reduced STSS mortality (low-certainty evidence)

A systematic review and meta-analysis (1 RCT + 40 observational studies, n = 1,918) found clindamycin treatment associated with lower mortality (OR 0.14, 95% CI 0.06–0.37) and, within clindamycin-treated patients, IVIG associated with lower mortality (OR 0.34, 95% CI 0.15–0.75) — both low-certainty. Age ≥ 65 increased mortality odds (OR 2.37, 95% CI 1.47–3.84): *"we found a statistically significant association between clindamycin treatment and mortality (n=144; OR 0.14, 95% CI 0.06 to 0.37)... Within clindamycin-treated STSS patients, we found a statistically significant association between intravenous Ig treatment and mortality (n=188; OR 0.34, 95% CI 0.15 to 0.75)"* [PMID: 36456018](#). A rigorous 2026 Swedish population-based cohort (n = 106) was, however, **inconclusive** for IVIG after adjusting for immortal-time bias: *"the hazard ratio was estimated at 1.69 (95% CI, .66-4.30). Although our study included a large population of patients with STSS, our results were inconclusive regarding the effect of IVIG"* [PMID: 41852552](#). The overall evidence therefore favors clindamycin more strongly than IVIG.

Intervention	Effect estimate	95% CI	Certainty	Source
Clindamycin → mortality	OR 0.14	0.06–0.37	Low	P 36456018
IVIG (in clindamycin-treated) → mortality	OR 0.34	0.15–0.75	Low	P 36456018
IVIG → 30-day mortality (adjusted)	HR 1.69	0.66–4.30	Inconclusive	P 41852552
Age ≥65 → mortality	OR 2.37	1.47–3.84	—	P 36456018

F004 — STSS epidemiology: rising post-COVID iGAS burden, emm1/M1UK predominance, high mortality

In a Korean nationwide iGAS cohort (2015–2024, n = 454), STSS accounted for 19.6% of iGAS and was the strongest independent predictor of mortality (aOR 20.07, 95% CI 9.30–43.30): *"streptococcal toxic shock syndrome (STSS, 19.6%)... were the most common iGAS infections. Intensive care unit admission was required in 28.9%... overall mortality was 15.5%. Mortality was independently associated with STSS (aOR 20.07, 95% CI 9.30-43.30"* [PMID: 42281793](#). A French post-COVID series (2022–2023) reported STSS in 24% of iGAS with 22% mortality and emm1 predominance: *"Streptococcal toxic shock syndrome (STSS) occurred in 24% of patients... Mortality rate was 22%... GAS emm1 genotype predominated (59%) and was involved in 79% of STSS"* [PMID: 42183688](#).

F005 — Mechanistic causal chain: superantigen cross-linking of MHC-II/TCR-Vβ drives the cytokine storm

Superantigens (SPE-A/C, SSA, SMEZ) simultaneously bind conserved regions of MHC class II on antigen-presenting cells and the TCR Vβ region, activating up to ~20% of the T-cell pool (versus ~0.01% for a conventional antigen), triggering a Th1/Th2/Th17/Th22 cytokine storm (TNF, IL-1, IL-2, IL-6, IFN-γ) within hours. Because they cross-link MHC-II, superantigens also activate APCs independently of T cells: *"the direct and simultaneous binding of SAGs with T-cell receptor (TCR)-bearing Vβ regions and conserved structures on major histocompatibility complex class II (MHC class II) on antigen-presenting cells (APCs) induces the activation of both cell types. However, by crosslinking MHC class II molecules, APCs can be activated by SAGs independently of*

T lymphocytes" PMID: 31412561. This sets in motion the toxic-shock cytokine storm: *"They target a large fraction of T cell pools to set in motion a 'cytokine storm' with severe and sometimes life-threatening consequences typically encountered in toxic shock syndrome"* PMID: 28632753. A subsequent immunosuppressive branch is driven by SPEA-induced regulatory T cells via TNF- α -TNFR2 signaling and monocyte PD-L1/kynurenine pathways: *"streptococcal pyrogenic exotoxin A (SPEA) promoted human CD4"* PMID: 40402252.

F006 — STSS clinical case definition and phenotype spectrum

STSS is operationally defined as **hypotension plus ≥ 2 of**: renal impairment, hepatic dysfunction, acute respiratory distress syndrome (ARDS), soft-tissue necrosis (e.g., necrotizing fasciitis), coagulopathy/DIC, and a generalized erythematous macular (sunburn-like) rash that may desquamate. Isolation of GAS from a sterile site defines a confirmed/definite case; isolation from a non-sterile site defines a probable case: *"The SSTS was defined as hypotension and at least 2 of these criteria: renal failure, hepatic failure, acute respiratory distress, tissue necrosis or desquamative erythematous rash"* PMID: 28705427. Fever is near-universal, and skin/soft-tissue lesions are the most frequent portal/manifestation.

Suggested HPO terms: HP:0001945 (Fever), HP:0002615 (Hypotension), HP:0000083 (Renal insufficiency), HP:0001410 (Decreased liver function), HP:0002098 (Respiratory distress), HP:0001928 (Abnormality of coagulation / DIC), HP:0000988 (Skin rash), HP:0007503 (Desquamation).

F007 — Animal models of STSS require HLA class II humanization; M protein + superantigen cooperate

Inbred mice are innately refractory to superantigen-mediated responses, so **HLA class II transgenic mice** (HLA-DR3, HLA-DR4, HLA-DQ) are required to recapitulate the human superantigen-driven cytokine storm. In HLA-transgenic mice infected with a lethal SpeC-expressing GAS isolate, both SpeC and M protein acting cooperatively were required for STSS; vaccination with the conserved M-protein peptide J8 (and passive anti-J8 + anti-SpeC antibodies) reduced bacterial burden and resolved disease: *"We modelled these in HLA-transgenic mice infected with a clinically lethal isolate expressing Streptococcal pyrogenic exotoxin (Spe) C and demonstrate that both SpeC and streptococcal M protein, acting cooperatively, are required for disease. Vaccination with a conserved M protein peptide, J8, protects against STSS"* PMID: 31517054. The rationale for humanized models: *"Inbred mice are*

innately refractive to SAg-mediated responses... the versatility of the HLA-II transgenic mouse model that allowed the biological validation of known genetic associations to GAS NSTI" PMID: 33079368.

F008 — Treatment backbone and the absence of a licensed GAS vaccine

Management of STSS with necrotizing soft-tissue infection requires early aggressive surgical debridement/complete removal of necrotic tissue (potentially amputation) plus directed antibiotics and hemodynamic support; delays increase mortality: *"Early surgical debridement with complete removal of necrotic tissue, including potential amputation, is essential to decrease mortality and other complications" PMID: 41417687.* The antibiotic regimen pairs a β -lactam (penicillin) for bactericidal killing with a protein-synthesis inhibitor (clindamycin or linezolid) to suppress toxin production, plus adjunctive IVIG. No GAS vaccine is licensed: *"developing an effective GAS vaccine has faced several challenges, including the complexity of GAS virulence mechanisms, the diversity of emm types, and the lack of suitable preclinical models" PMID: 40252837.*

Suggested NCIT terms: clindamycin (C376), linezolid (C1687), penicillin (C71634), intravenous immunoglobulin therapy (C603), surgical debridement (C15274).

F009 — Emergence of the toxigenic M1UK/M1T1 emm1 clone with elevated SpeA

A new emm1 lineage (M1UK) emerged in England, coinciding with unprecedented scarlet fever and invasive *S. pyogenes* activity; invasive emm1 isolates rose from 31% (2015) to 42% (2016). The lineage is defined by ~27 SNPs and characterized by **increased production of the superantigen SpeA**: *"Sequences of emm1 isolates from 2009-16 showed emergence of a new emm1 lineage (designated M1" PMID: 31519541.* Fatal STSS has been genomically linked to the M1(UK) (emm1/ST28) lineage carrying *slo*, *speC/speG*, *hasABC*, and *scpA*: *"S. pyogenes isolates from all sites were identified as M1(UK) (emm1/ST28) lineage... Key virulence genes included slo, speC/speG, hasABC and scpA" PMID: 41714973.*

F010 — Age-stratified risk factors; varicella + NSAID as a key pediatric gene-environment interaction

In children, principal predisposing factors are varicella (chickenpox; ~24–25%) and streptococcal pharyngotonsillitis, plus skin breaches; in adults, malignancy, IV drug use, diabetes mellitus, chronic wounds, and age >65 predominate: *"The main predisposing factors in*

children were varicella and streptococcal pharyngotonsillitis (25% and 19.8%, respectively), as opposed to malignancy, intravenous drug abuse and diabetes mellitus in adults (19.6%, 15.2% and 10.9%, respectively)" PMID: 23746128. A prospective case-control study established that NSAID use interacts with varicella to increase invasive GAS/NSTI risk: "To test the hypothesis that nonsteroidal antiinflammatory drug use increases the risk of necrotizing soft tissue infections and, secondarily, all invasive group A streptococcal (GAS) infections in children with primary varicella infection" PMID: 11331694.

F011 — Prognosis: high case-fatality, age/renal-failure prognostic factors, long-term sequelae

STSS carries the highest mortality among invasive GAS syndromes. In a 21-year Australian population data-linkage study (n = 933 iGAS hospitalizations), the hospitalized case-fatality rate was 5.2%, but most deaths were attributable to sepsis (80%), with necrotizing fasciitis (18%) and septic shock (13.3%) contributing; older age and renal failure were significant predictors: "Hospitalised case fatality rate was 5.2%. Intensive Care Unit (ICU) admission was required for 219 patients (23.5%)... Most deaths were attributable to sepsis (80%), with necrotising fasciitis (18%) and septic shock (13.3%) also contributing" PMID: 42684631. Survivors face substantial morbidity: "At discharge, 137/374 (37%) had not returned to premorbid function" PMID: 42004581. By analogy, survivors of childhood meningococcal septic shock experience high rates of skin scarring (48%) and orthopedic sequelae including amputation (8%) PMID: 19147623.

F012 — Diagnosis is clinical/microbiological; LRINEC and CT support necrotizing infection identification

STSS diagnosis rests on the clinical case definition plus isolation of GAS (culture/PCR) from a sterile (confirmed) or non-sterile (probable) site; there is no single confirmatory biomarker. Supporting laboratory abnormalities include leukocytosis with left shift, elevated CRP/procalcitonin, AKI (elevated creatinine), transaminitis, thrombocytopenia/DIC, elevated lactate, hypoalbuminemia, hypocalcemia, and creatine kinase elevation with myonecrosis. For associated necrotizing fasciitis, CT is the best imaging test and the LRINEC score aids risk stratification: "Computed tomography (CT) had sensitivity of 88.5% and specificity of 93.3%, while plain radiography had sensitivity of 48.9%... LRINEC ≥ 6 had sensitivity of 68.2% and specificity of 84.8%" PMID: 29672405. The LRINEC composition: "The score is determined by 6 serologic

markers: C-reactive protein (CRP), total white blood cell (WBC) count, hemoglobin, sodium, creatinine, and glucose" [PMID: 33570428](#). Surgical exploration/histopathology remains the diagnostic gold standard for necrotizing infection.

F013 — Naturally-occurring STSS/necrotizing fasciitis in dogs is caused by *Streptococcus canis* (group G)

A toxic-shock and necrotizing-fasciitis syndrome clinically analogous to human STSS occurs naturally in dogs, caused by *Streptococcus canis* (Lancefield group G; NCBITaxon:1114). In a prospective canine case series, dogs with streptococcal shock without NF died within 48 h, whereas those with NF survived after surgical debridement + antibiotics: "*3 dogs with streptococcal shock without necrotizing fasciitis died or were euthanatized within 48 hours of admission, whereas 4 dogs with streptococcal shock and necrotizing fasciitis survived following surgical debridement, supportive medical treatment, and treatment with antibiotics*" [PMID: 8870738](#). *S. canis* possesses M protein and streptolysin O but generally lacks most *S. pyogenes* superantigen genes: "*S. canis possesses M proteins and encodes streptolysin O, but lacks some of the other recognized virulence genes with significant homology to those in S. pyogenes*" [PMID: 10591501](#). A fluoroquinolone-inducible mitogen-encoding prophage may contribute: "*Both mitomycin and the fluoroquinolone enrofloxacin caused bacteriophage-induced lysis of S. canis strain 34, an isolate from a case of canine STSS and NF*" [PMID: 12761079](#) — cautioning that fluoroquinolone treatment could exacerbate canine disease.

F014 — Anatomical involvement: portal soft tissue plus systemic multi-organ injury via endothelium

STSS is a multi-system disorder. The primary portal/site is skin and soft tissue (subcutaneous tissue and superficial/deep fascia in necrotizing fasciitis; also pharynx, lung/pleura, muscle, postpartum uterus, joints). Systemically, the superantigen-driven cytokine storm targets the vascular endothelium (capillary leak → distributive/hypovolemic shock) and causes secondary failure of kidneys, liver, lungs (ARDS), heart (toxic cardiomyopathy), and the hematologic system (DIC). The multi-organ involvement is encoded in the case definition itself [PMID: 28705427](#), and the endothelium is the key target tissue: "*The slo-mediated endothelial lysis and speC/G-induced cytokine storm contributed to capillary leakage and rapid clinical deterioration*" [PMID: 41714973](#).

Suggested UBERON/CL terms: UBERON:0002097 (skin), UBERON:0002385 (fascia), UBERON:0001981 (blood vessel endothelium), UBERON:0002113 (kidney), UBERON:0002107 (liver), UBERON:0002048 (lung); CL:0000115 (endothelial cell), CL:0000624 (CD4+ T cell), CL:0000842 (antigen-presenting cell).

Detailed Section-by-Section Report

1. Disease Information

STSS is a rare, acute, toxin-mediated complication of invasive streptococcal infection defined by hypotension plus multi-organ dysfunction (F001, F006). **Key identifiers:** MONDO:0020544; MeSH "Shock, Septic" / "Streptococcal Infections"; ICD-10 A48.3 (Toxic shock syndrome); ICD-11 1B54. It has **no OMIM entry** because it is an acquired infectious disease, not a Mendelian condition. **Synonyms:** streptococcal toxic shock-like syndrome (STSLS/TSLs), streptococcal TSS, GAS toxic shock syndrome. Information is derived from **aggregated disease-level resources** (case series, cohort studies, meta-analyses) rather than individual EHR data.

2. Etiology

Causal factor: infectious — *Streptococcus pyogenes* (GAS) secreting superantigen exotoxins, with SLO cytolyisin (F001, F005, F009). Group C/G streptococci (*S. dysgalactiae* subsp. *equisimilis*) cause a minority of cases

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Genetic "risk": there are no patient causal genes; instead, host **HLA class II** genotype is the dominant severity modifier (F002). **Environmental/host risk factors** are age-stratified: children — varicella (~25%) and pharyngotonsillitis; adults — malignancy, IV drug use, diabetes, chronic wounds, age >65 (F010). **Gene-environment interaction:** NSAID use during primary varicella increases invasive GAS/NSTI risk in children (F010). **Protective factors:** naturally acquired anti-superantigen and anti-M-protein/conserved-antigen antibodies correlate with protection

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certain HLA class II haplotypes are relatively protective (F002).

3. Phenotypes

Core phenotypes (with type and typical frequency): fever (symptom; near-universal, HP:0001945), hypotension/shock (clinical sign; required for diagnosis, HP:0002615), sunburn-like erythematous rash with desquamation (physical manifestation, HP:0000988/HP:0007503), renal failure (organ, HP:0000083), hepatic dysfunction (HP:0001410), ARDS (HP:0002098), coagulopathy/DIC (HP:0001928), and soft-tissue necrosis/necrotizing fasciitis (~40% of cases). Onset is **acute (adult and pediatric)**; severity is **severe by definition**; progression is **rapid/fulminant** (multiorgan failure within hours). Quality-of-life impact among survivors is substantial (37% not returned to premorbid function at discharge; F011).

4. Genetic/Molecular Information

There are **no human causal genes, pathogenic variants, chromosomal abnormalities, or epigenetic lesions** in the patient (STSS is acquired). The relevant genetics are (a) the **host HLA class II locus** as a severity modifier (F002) and (b) the **bacterial virulence genome**: superantigen genes *speA*, *speC*, *speG*, *ssa*, *smez*; cytolysin *slo*; hyaluronic-acid capsule operon *hasABC*; C5a peptidase *scpA*; M-protein gene *emm* (F001, F009). The **emm1/M1UK** lineage overproduces SpeA and is defined by ~27 SNPs (F009).

5. Environmental Information

Infectious agent: *S. pyogenes* (NCBITaxon:1314), predominantly emm1/M1UK; less often group C/G streptococci. **Environmental/lifestyle contributors:** skin trauma or surgical wounds (portal of entry), IV drug use, NSAID use during varicella, and immunosuppression (F010). Respiratory viral co-infection is associated with more severe pediatric iGAS (

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6. Mechanism / Pathophysiology — Ordered Causal Chain

1. GAS (emm1/M1UK) colonizes/breaches skin, soft tissue, pharynx, or mucosa
| leads to
2. Local invasive infection (e.g., necrotizing fasciitis) + secretion of superantigens (SPE-A/C, SSA, SMEZ) and cytolyisin SLO
| leads to
3. Superantigens cross-link MHC class II (APC) directly to TCR-V β → activation of up to ~20% of the T-cell pool (bypasses antigen processing)
| branch: MHC-II cross-linking also activates APCs T-cell-independently
| results in
4. Polyclonal cytokine storm within hours (TNF, IL-1, IL-2, IL-6, IFN- γ , IL-17)
| in parallel: SLO lyses endothelial cells
| leads to
5. Vascular endothelial injury + capillary leak
| leads to
6. Distributive/hypovolemic shock (hypotension)
| leads to
7. Multi-organ dysfunction: AKI, hepatic failure, ARDS, DIC, toxic cardiomyopathy
| branch (late): SPEA-induced Tregs / monocyte PD-L1-kynurenine → immunoparalysis (impaired pathogen clearance)
| results in
8. STSS – high fever + shock + multi-organ failure (mortality ~20–45%)

Molecular pathways: TCR/MHC-II signaling, NF- κ B-driven pro-inflammatory cytokine transcription, IFN- γ -JAK-STAT1 (small-intestinal immunopathology is a major driver of lethality in HLA-DR3 CRS models; ruxolitinib is protective —).

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Cellular processes: polyclonal T-cell activation, APC activation, MAIT-cell hyperactivation then anergy

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endothelial cytolysis, neutrophil-driven injury, immunothrombosis, apoptosis/necrosis

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Upstream: superantigen/SLO release; **downstream:** capillary leak, shock, organ failure, immunoparalysis. **GO terms:** GO:0042110 (T-cell activation), GO:0006954 (inflammatory response), GO:0050729 (positive regulation of inflammatory response), GO:0002827 (positive regulation of Th1 immune response). **CL terms:** CL:0000624 (CD4⁺ T cell), CL:0000940 (MAIT cell), CL:0000842 (APC/monocyte), CL:0000115 (endothelial cell).

7. Anatomical Structures Affected

Primary: skin and soft tissue (subcutaneous tissue, superficial/deep fascia; UBERON:0002097, UBERON:0002385), pharynx, lung/pleura, muscle, postpartum uterus, joints. **Secondary/systemic:** vascular endothelium (key mediator; UBERON:0001981), kidney, liver, lung, heart, hematologic system (F014). **Body systems:** integumentary, cardiovascular, renal, hepatic, respiratory, hematologic. **Lateralization:** soft-tissue focus typically unilateral/localized (e.g., a limb); shock and organ failure are systemic/bilateral.

8. Temporal Development

Onset: acute, at any age (bimodal risk: young children with varicella; adults >65). **Progression:** fulminant — shock and multiorgan failure within hours (F004, F006). **Course:** self-limited if survived (not chronic); no relapsing-remitting pattern. **Critical period:** the first hours are the decisive window for source control, antibiotics, and IVIG; delays sharply increase mortality (F008).

9. Inheritance and Population

Not heritable (acquired infection). **Epidemiology:** STSS constitutes ~19.6–24% of iGAS (F004). iGAS ICU admission ~24–29%; overall iGAS mortality ~14–16%, but STSS-specific mortality is markedly higher (~20–45%). **Population demographics:** bimodal age distribution; adults >65 at highest mortality risk; both sexes affected. **Geographic/temporal:** post-COVID surge across Europe, Korea, Australia, driven by emm1/M1UK clonal expansion (F004, F009). The relevant "founder"/clonal effect is bacterial (M1UK), not human.

10. Diagnostics

Clinical/microbiological (F012): clinical case definition (hypotension + ≥ 2 organ criteria) plus GAS isolation from sterile (confirmed) or non-sterile (probable) site. **Labs:** leukocytosis with left shift, $\uparrow$ CRP/procalcitonin, $\uparrow$ creatinine (AKI), transaminitis, thrombocytopenia/DIC, $\uparrow$ lactate, hypoalbuminemia, hypocalcemia, $\uparrow$ CK. **Imaging:** CT is best for necrotizing infection (sensitivity 88.5%, specificity 93.3%). **LRINEC score** (CRP, WBC, hemoglobin, sodium, creatinine, glucose) ≥ 6 supports NF (specific but insensitive). **Strain characterization:** blood/wound/tissue cultures, emm typing (Rt-PCR or sequencing), whole-genome sequencing. **Gold standard for NF:** surgical exploration/histopathology. Patient genetic testing is **not applicable**.

11. Outcome/Prognosis

STSS has the **highest mortality of invasive GAS syndromes** (F011). Population case-fatality for hospitalized iGAS is ~5%, but STSS-associated cohorts report ~20–45%. **Prognostic factors:** age ≥ 65 (OR 2.37), renal failure, elevated lactate, delayed source control. **Complications:** septic shock, necrotizing fasciitis, DIC, multi-organ failure. **Morbidity in survivors:** amputation, skin scarring, limb-length discrepancy, functional impairment (37% not returned to premorbid function at discharge; F011).

12. Treatment

Three-pillar strategy (F003, F008): 1. **Source control:** early aggressive surgical debridement (potentially amputation) for necrotizing infection — essential to reduce mortality (NCIT: surgical debridement C15274). 2. **Antibiotics:** β -lactam (penicillin, NCIT C71634) for bactericidal killing PLUS a protein-synthesis inhibitor — clindamycin (C376) or linezolid (C1687) — to suppress toxin production (the "Eagle effect" rationale). 3. **Adjuncts:** IVIG (C603) to neutralize superantigens (low-certainty benefit; inconclusive in the most rigorous cohort), corticosteroids, and intensive organ support (vasopressors, mechanical ventilation, renal replacement, occasionally ECMO/extracorporeal immunomodulation —

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Experimental/future: JAK inhibition (ruxolitinib) protective in CRS models (

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IgM/IgA-enriched immunoglobulins

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vaccine candidates (J8/StreptInCor conserved M-protein peptides, Slr, non-M conserved antigens). **No licensed GAS vaccine** (F008). **Pharmacogenomics:** no established patient PGx; HLA class II modulates response magnitude (F002).

13. Prevention

Primary: prompt treatment of GAS pharyngitis/skin infection; varicella vaccination (removes a major pediatric risk factor); avoiding NSAIDs during varicella; wound care. **Secondary:** early recognition and empiric therapy in at-risk patients; post-exposure chemoprophylaxis for close household contacts of severe iGAS (per public-health guidance). **Tertiary:** aggressive source

control and toxin suppression to prevent progression. **Immunization:** no licensed GAS vaccine; multiple candidates in development (F008). **Public health:** iGAS surveillance, emm typing, outbreak investigation. **Counseling:** genetic counseling not applicable (non-heritable).

14. Other Species / Natural Disease

A **naturally-occurring canine STSS/necrotizing fasciitis** exists, caused by *Streptococcus canis* (Lancefield group G; NCBITaxon:1114) (F013). Canine *S. canis* possesses M protein and SLO but lacks most *S. pyogenes* superantigen genes, and canine STSS is largely **non-clonal** (unlike human M1 clonal expansion;

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A fluoroquinolone-inducible mitogen-encoding prophage (phiSC1, gene *scm*) may contribute superantigen-like activity, cautioning against fluoroquinolone use in canine cases (F013). This provides a comparative-pathology model illustrating that M protein + SLO can drive shock/NF even with a divergent superantigen repertoire. Group C/G streptococci also cause STSS-like disease in humans

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15. Model Organisms

The essential model is the **HLA class II transgenic mouse** (HLA-DR3, HLA-DR4, HLA-DQ), because inbred mice are innately refractory to superantigens (F007). These models recapitulate the superantigen-driven cytokine storm and demonstrated the cooperative requirement of SpeC + M protein and J8-vaccine protection

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Advanced recombinant inbred BXD mice map host genetic loci modulating GAS necrotizing soft-tissue infection

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HLA-DR3 cytokine-release-syndrome models revealed small-intestinal immunopathology and IFN- γ /IL-17A/JAK signaling as pathogenic

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Limitations: murine models incompletely capture human hemodynamics and chronic sequelae; superantigen refractoriness necessitates humanization. **Resources:** MGI, IMSR for HLA-transgenic strains.

Mechanistic Model / Interpretation

The unifying model is that **STSS is a host-response disease triggered by bacterial superantigens acting on a genetically permissive immune system**. Two converging axes cause shock:

Axis	Effector	Immediate effect	Downstream
Immunological	SPE-A/C, SSA, SMEZ superantigens	MHC-II × TCR-V β cross-link → polyclonal T-cell + APC activation	Cytokine storm (TNF, IL-1/2/6, IFN- γ , IL-17) → capillary leak
Cytolytic	Streptolysin O (SLO)	Direct endothelial lysis	Capillary leak, tissue necrosis

Both axes converge on **endothelial dysfunction → capillary leak → distributive shock → multi-organ failure**. Host **HLA class II** genotype sets the gain on the immunological axis, explaining why the same clone produces divergent outcomes. The bacterial side is amplified when a hyper-toxigenic clone (**M1UK**, over-producing SpeA) circulates. A late **immunoparalysis** branch (SPEA-induced Tregs, monocyte PD-L1/kynurenine, MAIT anergy) impairs pathogen clearance and may worsen secondary infection. This model directly rationalizes therapy: kill the organism (β -lactam), silence the toxin factory (clindamycin/linezolid), neutralize circulating superantigen (IVIg), and physically remove the toxin source (surgical debridement).

Evidence Base

PMID	Role	How it supports the findings
39886786	Case report	Defines the STSS triad and GAS causation (F001)
41714973	Genomic case	SLO endothelial lysis + SPE cytokine storm; M1(UK) virulence genes (F001, F009, F014)
14620152	Review	HLA class II determines severity (F002)
36456018	Meta-analysis	Clindamycin/IVIG mortality associations; age ≥ 65 (F003)
41852552	Cohort	IVIG inconclusive after bias adjustment (F003)
42281793	Nationwide cohort	STSS 19.6% of iGAS; aOR 20 for mortality (F004)
42183688	Series	Post-COVID STSS 24%, 22% mortality, emm1 (F004)
31412561	Mechanism	MHC-II/TCR-V β cross-link; T-cell-independent APC activation (F005)
28632753	Mechanism	Superantigen cytokine storm; MAIT anergy (F005)
40402252	Mechanism	SPEA-induced Tregs (immunosuppression branch) (F005)
28705427	Series	Clinical case definition/organ criteria (F006, F014)
31517054	Model	HLA-humanized mouse; SpeC+M cooperation; J8 vaccine (F007)
33079368	Model	Rationale for HLA-II transgenic models (F007)
41417687	Position statement	Early debridement essential (F008)
40252837	Review	No licensed GAS vaccine; barriers (F008)
31519541	Molecular epidemiology	M1UK emergence, elevated SpeA (F009)
23746128	Epidemiology	Age-stratified risk factors (F010)
11331694	Case-control	NSAID $\times$ varicella interaction (F010)
42684631	Data linkage	iGAS case-fatality, causes of death (F011)
42004581	Cohort	37% not returned to premorbid function (F011)
19147623	Cohort	Long-term scarring/orthopedic sequelae of septic shock (F011)
29672405	Meta-analysis	CT/LRINEC diagnostic accuracy (F012)

PMID	Role	How it supports the findings
33570428	Review	LRINEC composition (F012)
8870738	Veterinary series	Canine STSS/NF outcomes (F013)
10591501	Comparative genomics	<i>S. canis</i> virulence repertoire (F013)
12761079	Molecular	Fluoroquinolone-inducible mitogen prophage (F013)
32676080	Model	JAK-STAT/IFN- γ intestinal immunopathology; ruxolitinib
40781379	Serology	Natural protective humoral immunity (protective factors)

Note on citation flags: Several citations (PMIDs 42183688, 41714973, 42684631) were flagged during verification as "mismatch" — the quoted text is consistent with the finding but the automated snippet-to-abstract match was imperfect (abstracts were partially truncated in the source). These claims are supported but should be re-verified against full-text abstracts before database ingestion.

Limitations and Knowledge Gaps

- 1. Therapeutic uncertainty (IVIG).** The mortality benefit of IVIG is low-certainty and became **inconclusive** in the most rigorous, bias-adjusted cohort (HR 1.69, 95% CI 0.66–4.30; [P 41852552](#)). A definitive adequately powered RCT is lacking. Clindamycin's benefit, while consistent, also rests largely on observational data.
- 2. Host genetics not resolved to variant level.** HLA class II is established as a severity modifier, but specific protective/risk alleles and their effect sizes for STSS are not fully catalogued; no GWAS-scale analysis of STSS outcomes was identified.
- 3. Biomarker gap.** There is no single confirmatory diagnostic biomarker; LRINEC is insensitive; early distinction of STSS from other septic shock relies on clinical suspicion and culture.

4. **Immunoparalysis under-characterized.** The late immunosuppressive branch (Tregs, PD-L1/kynurenine, MAIT anergy) is mechanistically described but its clinical prognostic and therapeutic implications remain investigational.
5. **Model limitations.** HLA-transgenic mice are essential but incompletely capture human hemodynamics, chronic sequelae, and polymicrobial context.
6. **Vaccine absence.** emm-type diversity, virulence complexity, and preclinical model gaps continue to block a licensed vaccine.
7. **Citation verification.** Some abstracts were truncated; a subset of quotes need full-text re-verification (see note above).

Proposed Follow-up Experiments / Actions

1. **Definitive IVIG RCT / emulated target trial** in STSS with pre-specified adjustment for immortal-time bias, stratified by superantigen genotype and HLA class II type, to resolve the clindamycin-adjacent IVIG question.
2. **Host-genetics study:** targeted HLA class II typing (and broader immunogenetic panels) in a large STSS-vs-non-severe-iGAS cohort to define specific risk/protective alleles and effect sizes for knowledge-base annotation.
3. **Prospective biomarker-stratified cohort** (per

P 42459336

 using monocytic HLA-DR expression, lymphocyte indices, and cytokine trajectories to develop a diagnostic/prognostic panel and phenotype the hyperinflammation→immunoparalysis transition.
4. **JAK-inhibitor translational study:** evaluate ruxolitinib (or selective JAK1/2 inhibition) as host-directed adjunct, building on HLA-DR3 CRS model protection
 (

P 32676080

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 with careful attention to infection-clearance risk.
5. **Genomic surveillance** of emm1/M1UK and emerging lineages (including group C/G SDSE) integrated with clinical outcomes to track toxigenic clonal shifts driving STSS incidence.
6. **Vaccine advancement:** progress conserved-antigen candidates (J8/StreptInCor, Slr, non-M conserved antigens; PMIDs 40252837, 42298213, 40781379) through HLA-humanized preclinical models toward Phase I.

7. **Comparative one-health study** of canine *S. canis* STSS to test whether M protein + SLO alone suffice for shock, informing minimal virulence requirements and cautioning on fluoroquinolone use

(P 12761079).

Report compiled from 14 confirmed findings and 66 reviewed papers across 5 investigation iterations. Evidence source types span human clinical cohorts, meta-analyses, HLA-humanized model organisms, in vitro mechanistic studies, and veterinary comparative pathology.

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